Hyperspectral Image Restoration With Self-Supervised Learning: A Two-Stage Training Approach

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***Abstract*— Hyperspectral image (HSI) denoising is a crucial preprocessing task to improve the performance of the subsequent HSI interpretation and applications. With recent progress in deep learning, HSI denoising methods based on deep neural networks have attracted increasing interest and achieved the state-of-the-art performance. Nevertheless, most of these methods are based on network structures originally developed for grayscale and color images and require a change of network structure to be applicable to HSIs. The new network architectures often lead to complicated models and limited flexibility, which, in turn, result in difficulty in learning and demand of a large number of training samples. In this article, we propose an innovative two-stage learning method including pretraining and fine-tuning procedures. In the first stage, a denoising convolutional neural network can be pretrained with pairs of corrupted and clean images. In the second stage, the pretrained network is fine-tuned via a self-supervised learning strategy to capture the spectral correlation in HSIs. The training pairs in the second stage are constructed from the neighboring band images in the target noisy HSI, leading to a novel idea of embedding spectral information into denoiser through the target image, rather than the change of the network architecture. This model has strong adaptability such that many image denoising networks can be easily adopted for HSIs, while the external hyperspectral training set is optional but not mandatory. Experimental results show that our method has competitive performance compared with the state-of-the-art approaches.**

***Index Terms*— Deep neural networks (DNNs), denoising, hyperspectral images (HSIs), self-supervised learning.**

I. Introduction

HYPERSPECTRAL sensors collect information across the electromagnetic spectrum in many contiguous and narrow wavelength intervals, producing hyperspectral images (HSIs) with hundreds of spectral bands [1]. Compared with other types of images, such as grayscale, color, and multispectral images, HSIs are rich in spectral information, making the technology widely used for scene classification [2]–[4], anomaly detection [5], [6], object tracking [7], [8], among many others. However, due to the limited sensitivity of sensors, low light intensity for each band, and environmental interference, the captured HSIs are easily corrupted by noises that affect the subsequent applications. Therefore, HSI denoising is an important task.

Traditional model-based HSI denoising methods normally process HSIs band-by-band or in a joint spectral–spatial manner. In band-by-band HSI denoising, many image denoising algorithms developed for grayscale images can be directly applied, such as wavelet transformation, K-singular value decomposition (K-SVD), nonlocal means (NLMs) [9], block matching 3-D (BM3D) [10], and more. Nevertheless, band-by-band methods neglect the intrinsic spectral correlation in an HSI, leading to a spectral discontinuity in restored HSI. As an alternative, joint spectral–spatial methods treat HSI as a cube, therefore, the 3-D structure is taken into account. For example, low-rank recovery of a matrix [11] can be extended to the low-rank recovery of a three-order tensor [12]–[15]. Similarly, sparse coding-based methods can also be extended from operations on 2-D patches [16] to 3-D patches [17]–[19]. Both extensions aim to consider the joint spectral–spatial correlation between neighboring spectral bands and spatial locations. Although these model-based methods work well when the prior model assumptions match well with the HSIs to be processed, they are normally computationally expensive and require parameter tuning on different HSIs.

Recently, deep learning has achieved great success in image processing. On image restoration tasks, deep neural network (DNN)-based methods directly learn a nonlinear mapping from corrupted to restored images through noise-clean image pairs; therefore, the denoising problem can be solved in data-driven end-to-end manner. Most DNN approaches are based on convolutional neural networks (CNNs) due to their large modeling capacity and remarkable advances in network training and architecture design. For example, feedforward denoising CNNs (DnCNN) [20] performs blind denoising under unknown noise levels. It is recognized as a gold standard of DNN-based image restoration methods on grayscale and color images, uses residual learning and batch normalization to speed up the training process and boost the denoising

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performance. FFDNet [21] is another common used CNN for image denoising. With a tunable noise level map as the input, FFDNet is able to handle a wide range of noise levels effectively with a single network. By working on downsampled subimages, FFDNet achieves a good tradeoff between inference speed and denoising performance. Compared with model-based image restoration methods, data-driven DNNs are easier to use thanks to their general architectures without relying on domain knowledge on denoising problems. Although the training of DNNs is time-consuming, the denoising procedure is fast as only one-time feedforward is required.

When applied on HSIs, these ordinary DNN denoisers process images band-by-band so that the spectral correlations of bands are also ignored. To address this problem, several recent DNN methods are specially designed for HSI restoration [22]–[27]. For example, separable 3-D CNN (SP3DCNN) [25] utilizes 3-D convolution instead of 2-D convolution to combine spatial and spectral information. In order to reduce the number of parameters, the 3-D convolutional filtering is decomposed into a concatenation of 2-D spatial filtering and 1-D spectral filtering, and this simplification does not visibly degrade the denoising performance. HSI denoising with a deep CNN (HSID-CNN) [26] is based on a residual CNN but contains two inputs: a noise image of a certain band and a cube composed of adjacent images of that band in order to take both spatial and spectral information into consideration. Features are extracted from two inputs separately, and finally, be combined to obtain the denoised image of the input band after a series of convolutional operators. However, HSID-CNN requires training of a specific network for each level of noise in the data and does not provide a global solution to the denoising problem [27]. HSI-SDeCNN [27] improves HSID-CNN from two aspects: a downsampling operation is performed before feature extraction to reduce the computational overhead and, a noise-level map is added as additional input information to make the network flexible and adaptive to multilevel noises, without the need to train different models for different noise levels.

Different from the ordinary DNNs for grayscale or RGB color image restoration, the aforementioned DNN-based HSI denoisers have specific network architectures to fully exploit the rich spectral information contained in HSIs. However, embedding spectral information into network architecture requires more complex networks, leading to more difficult learning and worse generalization. Moreover, larger HSI training sets are required to train more complex networks. These problems limit the adaptability, scalability, and generalization of DNN models in practical HSI applications.

In this article, we propose a novel solution that does not require a specific design of network architecture for HSIs. In other words, we directly use an ordinary DNN-based 2-D image denoiser as the backbone model. The spectral information of HSI is considered in the training procedure, rather than by extending network architecture. Specifically, our network learning procedure includes two stages: pretraining and fine-tuning. The pretraining stage learns a selected DNN originally developed for grayscale or color images. The pretrained network can be applied to noisy HSIs in a band-by-band manner. To capture the spectral information, the fine-tuning stage further trains the pretrained network by a self-supervised learning scheme, i.e., the corrupted HSI to be restored is used to build the training set. Our fine-tuning method has two outstanding advantages. First, the training samples are generated from the target corrupted HSI, enabling the learned network to fit the target corrupted HSI. This is different from most DNN-based denoising methods for which training samples and the corrupted image to be restored are different; therefore, the discrepancy between training and testing samples can be overcome. Second, the spectral information between neighboring bands is reflected in the training pairs via self-supervised learning, avoiding designing a specific network architecture for HSIs. Inspired by noise2noise [28] learning theory, we generate the noise-noise training pairs between any two neighboring/adjacent bands in the corrupted HSI. This is based on the assumption that the neighboring band images in an HSI could be considered as the independent corrupted images from the same underlying clean image. This method embeds the spectral information in the training pairs, which is different from most spectral–spatial DNNs in which the spectral information is reflected in the input of the network so that the network architecture is not necessary to be redesigned to tackle spectral–spatial input.

The main contributions of our method can be summarized as follows.

1) Two-stage learning paradigm for HSI denoising is proposed. The pretraining stage uses any training samples from grayscale, color, multispectral, and HSI image sets to get the initial parameters of the network. Then, the fine-tuning stage uses noise-noise training samples obtained from a corrupted HSI to be restored. This paradigm makes full use of a large training set in pretraining and a small but more targeted training set in fine-tuning. It reduces the difficulty of training data acquisition, in particular, external HSI training data are not mandatory. As we know, combining fine-tuning and self-supervised schemes together is not proposed before, at least in the field of HSI restoration.

2) Unlike other DNN-based HSI denoising methods, we rely on the training samples, rather than network architecture to embed the spectral information into the learning procedure. This makes our method more general and simpler, leading to an innovative, effective, and general technique for DNN-based HSI denoising, which allows any 2-D image-based denoising network used in this framework.

3) The extensive experiments on both synthetic and real-world data show the competitive performance compared with the state-of-the-art model-based and DNN-based denoising approaches.

The remainder of this article is organized as follows. In Section II, we introduce the preliminaries of our method, including noise modeling, DnCNN [20], and the noise-to-noise learning theory [28]. In Section III, our two-stage scheme is described, and the relation to some self-supervised learning-based methods is also discussed. The experimental results and comparisons with the state-of-the-art denoising methods

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are given in Section IV. Finally, we conclude our work in Section V.

II. Preliminaries

In this section, we first give the HSI denoising model and then introduce DnCNN and noise-to-noise learning theory as the foundation of our work. DnCNN is selected as the DNN model for image restoration, nonetheless, many other DNN-based image denoiser can also be chosen instead, as our method does not require a new network architecture specifically designed for HSI restoration. Noise-to-noise learning is the theoretical basis of training sample generation in the fine-tuning stage.

A. HSI Denoising Model

The quality of an HSI is dependent on several factors. Especially, there are two main sources of noises in HSIs, the first is from the imaging device itself, such as electromagnetic interference and the second comes from the observation environment. For example, the noise types include photon (or shot) noise, excess low-frequency noise, thermal noise, and stripe noise due to mechanical scanning motion. These different noise types have been modeled as the corresponding statistical distributions by mathematical simplification. The common noise distributions studied in the HSI literature are Gaussian noise, impulse noise, strips, deadlines, and Poisson noise.

Given an HSI $\mathcal{X}$ to be denoised, it can be considered as a corrupted version of the underlying clean HSI $\mathcal{Y}$. If the noise is additive, we have

$$\mathcal{X} = \mathcal{Y} + \mathcal{N} \tag{1}$$

where $\mathcal{N}$ represents noise. Some multiplicative noises, such as Poisson noise, can be transformed to additive noise [29]. In this article, we mainly discuss Gaussian noise and impulse noise, and both are additive. Furthermore, we assume that the noises in a band image are independent, i.e., they are independent in spatial space. Gaussian and impulse noises nearly satisfy this assumption. In spectral space of HSI, the signal correlation across the band is one of the crucial sources used for spectral denoising or spectral–spatial denoising [30], but the noise correlation across the band is always ignored and assumed to be independent.

However, these assumptions of noise are just to simplify the theoretic analysis. In fact, our proposed method works well in practice, which can be proved by the experimental results on real-world HSI datasets corrupted by various types of noise in Sections IV-D and IV-E.

B. DnCNN

We choose DnCNN [20] because it is a common benchmark for DNN-based image denoisers, which uses residual learning for blind denoising. Different from most image denoising networks that usually train a specific model for Gaussian noise at a certain noise level, DnCNN is able to tackle Gaussian noise with unknown noise levels by the residual learning paradigm. DnCNN implicitly removes the latent clean image in the hidden layers so that the mapping from the noisy input to the clean output is close to the identity mapping, and this residual mapping is easy to optimize [31]. Moreover, residual learning and batch normalization can benefit from each other, and boost the denoising performance.

When applied to a corrupted image with unknown noise, DnCNN does not need to estimate the noise level and train the model with the estimated noise level. Instead, it uses corrupted images with a wide range of noise levels to train a single DnCNN model, and the learned DnCNN can be utilized to restore a corrupted image within the noise level range of training samples. The blind denoising of DnCNN is very useful for band-by-band HSI denoising, as in general different band images tend to have different noise levels. Therefore, only a single DnCNN is required to be trained for HSI denoising, which is convenient for real applications.

DnCNN has a CNN architecture similar to VGG network [32] but is modified for image restoration. In our experiments, 17 convolutional layers are used. The first layer consists of convolutions and subsequent ReLU activation [33], which outputs a 64-D feature map given 64 3×3 convolutional kernels. The next 15 layers are repeated structures, each containing 64 3×3 convolutions followed by batch normalization and ReLU activation [34]. The last layer is a simple 3×3 convolutional layer. The training pairs for DnCNN are usually noise-clean corresponding images, and they are the one-channel images obtained from grayscale, color, and multi/HSI images, as both input and output of DnCNN in our experiments are one-channel images. Although DnCNN can tackle RGB color images by considering the spectral correlation between RGB bands, only considering three bands are not sufficient for HSIs; therefore, here we do not consider the DnCNN model for RGB color images.

C. Noise-to-Noise Learning

In general, a deep denoising network learns a mapping from a corrupted observation to the corresponding clean version by a number of training pairs of noise-clean images. Specifically, constructing a training dataset, including pairs $(\widetilde{x}_i, y_i)$ of corrupted inputs $\widetilde{x}_i$ and the corresponding clean targets y_i, the following minimization of empirical risk shall be achieved:

$$\arg\min_{\theta} \sum_{i} L\left(f_\theta(\widetilde{x}_i), y_i\right) \tag{2}$$

where L is the loss function, f_θ is the mapping function, and θ represents the parameters of the network. It is hard to obtain a large training set in practice [35], especially when the training set shall cover the same data distribution of the testing samples.

This issue is tackled by self-supervised learning recently proposed for denoising, such as noise2noise [28], noise2self [36], and noise2void [37], which learn to restore images by only looking at corrupted examples instead of requiring a large number of noisy-clean training pairs. This strategy greatly improves the feasibility of DNN-based image denoisers.

In our method, noise-to-noise learning theory [28] is adopted. This self-supervised learning method is based on a simple and powerful conclusion that adding zero-mean Gaussian noise to a clean image to build the training targets

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will not change the content of network learning. This means that it is possible to train a denoising network using pairs of noisy images $(\widetilde{x}, \widetilde{y})$, as long as two images in a pair have the same underlying clean image y and their noises are independent of each other.

If we consider the training of an image restoration network as a point estimation procedure, (2) can be rewritten as

$$\arg\min_{\theta} \ \mathbb{E}_{(\widetilde{x},y)}\{L(f_\theta(\widetilde{x}), y)\}. \tag{3}$$

It minimizes the expectation of the loss. If L_2 loss is used, i.e., $L(f_\theta(\widetilde{x}), y) = ||f_\theta(\widetilde{x}) - y)||_2^2$, the minimum is found at the arithmetic mean of the observations

$$f_\theta(\widetilde{x}) = \mathbb{E}_y(y). \tag{4}$$

In other words, the network learns to output the average of all input images, which usually results in a blurry predicted image, but not the desired solution [38].

Alternatively, if we add noise to the clean image to generate a group of target images, and make their expectation equal to the original clean image, the estimate becomes the clean image according to (4). Combined with (2), a new objective function can be obtained

$$\arg\min_{\theta} \sum_i L\left(f_\theta(\widetilde{x_i}), \widetilde{y_i}\right) \tag{5}$$

where both inputs $\widetilde{x_i}$ and targets $\widetilde{y_i}$ are corrupted images with independent zero-mean Gaussian noise. In theory, given infinite data, the solution of (5) is the same as (2). This suggests the strategy of generating training sets for noise-to-noise learning strategy: collecting a training set, including many groups of noise-noise image pairs, and all pairs in the same group are corrupted by Gaussian noise from an identical clean image. It is proven [28] that this strategy well approximates the ideal solution in many cases and achieves a good denoising performance close to using noise-to-clean training data. The experimental results in this article also show its favorable properties.

III. Proposed Two-Stage Denoising Approach

In this section, we propose a two-stage denoising scheme based on 2-D image denoising networks and explain how it can be applied to the HSI denoising task. Moreover, we will discuss its distinctive properties from related works in HSI restoration based on self-supervised learning.

A. Description of Method

Many DNN-based HSI restoration methods use 3-D HSIs or their patches as training samples, but it is more difficult to collect HSI training pairs than grayscale or color image pairs. Furthermore, the distribution of 3-D images is much more complex than 2-D images, leading to more required training samples. In addition, there are a number of associated problems of 3-D learning, e.g., the number of spectral bands must be adjusted to be the same between training and test HSIs, and the noisy levels always need to be similar among bands. On the contrary, a remarkable advantage of single-channel (2-D image/patch-based) DNNs is that the training pairs can be obtained from all types of optical images including grayscale, color, and multi/hyperspectral, as their individual spectral band image forms a training sample independently. It makes the collection of training samples relatively easy.

Our two-stage training procedure contains pretraining and fine-tuning. In the pretraining stage, an initial single-channel DnCNN is learned using an external training set of noisy-clean image pairs. For HSI denoising, the initial DnCNN does not consider the correlations among spectral bands. Furthermore, the training samples used for pretraining may not match the HSIs in terms of scene, data distribution, imaging condition, and so on.

Inspired by noise-to-noise learning theory, in the second stage, a new fine-tuning strategy based on self-supervised learning is proposed, in which the target HSI to be denoised is used as the source of generating noise-noise training pairs. In other words, the HSI takes the roles of both training and testing at the same time during the fine-tuning procedure. Image restoration without clean data assumes it is possible to learn a mapping from corrupted observations to clean signals by only looking at corrupted examples. Each training pair is composed of two corrupted images with noises independently added to the same target.

To meet this requirement, we establish the training pairs through any two neighboring/adjacent bands, i.e., a pair consists of two neighboring band images in a corrupted HSI, according to the assumption that the neighboring band images in an HSI can be considered as the independent corrupted images from the same underlying clean image. This assumption is based on the mechanism of hyperspectral imaging that all spectral band images in an HSI are captured from the same scene, and their spectrum ranges are continuous. This is similar to video restoration [39], in which consecutive frames are adopted as observations of the similar underlying clean image after motion calibration. This training sample collection method takes advantage of the correlation (redundancy) of the spectral bands of HSIs to remedy the drawback of single-channel DnCNN in that the spectral information is ignored.

Given an HSI $\mathcal{X}$ to be denoised, its size is $H \times W \times B$, where B denotes the number of bands, and $H \times W$ is the size of each band image. The ith band image is denoted as $\widetilde{x_i}, i = 1, \ldots, B$. The neighboring band images are used to construct the training pairs that approximately satisfy the conditions of two images in a pair being noise-independent and having the same underlying clean image. A pair of training samples can be formed as

$$\left(\widetilde{x_i}, \widetilde{x_j}\right) \quad i, j \in [1, B], \quad 0 < |i - j| \leq k \tag{6}$$

where k represents the size of neighbor, which means that for a certain band i, bands in the range from $i - k$ to $i + k$ (not including i itself) can be used to build the training pairs with the ith band. The value selection of k is dependent on how close two bands should be so that they can be assumed as the noisy samples corrupted by independent noise on the same underlying clean image. With the large value of k, more training pairs can be generated from an HSI,

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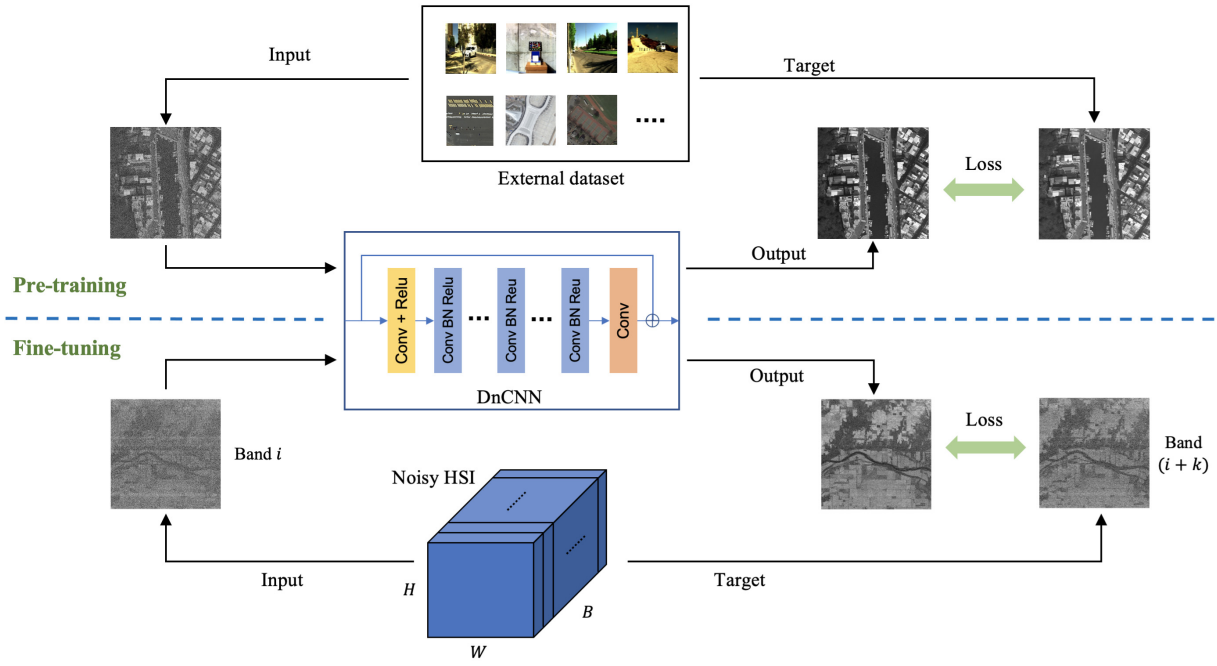


Fig. 1. Flowchart of the proposed two-stage learning method for HSI restoration.

while the assumption of having the same underlying clean image becomes to be harder to satisfy. The value selection of k is related to the imaging device and environment, but our experimental results in Section IV suggest that $k = 3$ is suitable to almost all cases.

After the training set is constructed, the loss of fine-tuning is

$$\arg\min_{\theta} \sum_{i=1}^{B} \sum_{j \neq i} L\left(f_\theta(\widetilde{x}_i), \widetilde{x}_j\right) \quad j \in [\max(i-k,1), \min(i+k,B)] \tag{7}$$

where L denotes the error measurement function.

L_2 norm is chosen as the error function as this norm processes many desirable properties [40], such as leading to stable solution and preventing overfitting. However, L_2 suffers from some well-known limitations, including the Gaussian noise assumption and that the impact of noise is independent of the local characteristics of the image, which are not always valid. From the view of network training, L_2 penalizes larger errors, but is more tolerant to small errors, regardless of the underlying structure in the image. On the other hand, L_1 does not over-penalize larger errors and consequently, it has different convergence properties from L_2. In image restoration, L_2 is often used to handle Gaussian noise, while L_1 loss may be able to better handle other complex noise, such as impulse noise, mixed noise, and so on [41].

Therefore, in our method, we use a mixed loss combining both L_1 norm and L_2 norm error measures

$$\begin{aligned} L(x,y) &= \alpha L_1(x,y) + (1-\alpha) L_2(x,y) \\ &= \alpha ||x-y||_1 + (1-\alpha)||x-y||_2^2 \end{aligned} \tag{8}$$

where α is a hyperparameter to control the weights of Gaussian noise and other noises in images, which we will discuss in Section IV.

When only L_1 loss is used, the minimum of (7) is found at the median of the observations

$$f_\theta(\widetilde{x}) = \text{Median}_{\widetilde{x}}(\widetilde{x}). \tag{9}$$

In other words, the network learns to output the median of all input images. If the impulse noise is independent in both spatial and spectral spaces, L_1 loss usually results in desired clean band images. Therefore, the noise-to-noise training strategy still can be used for impulse noise. In practice, a mixed loss with L_1 and L_2 norms always generates better HSI restoration results, as real HSIs have various types of noise.

Since the fine-tuning stage is based on self-supervised learning, it allows us to train and test the network at the same time. Here, the ADAM algorithm is used for optimization. After convergence, we stop training and use the tuned network to predict the denoised HSI. The input is a corrupted band image, and the corresponding output is the denoised image. The final denoised HSI is obtained after all denoised band images are predicted in a band-by-band manner. The whole procedure of our method is shown in Fig. 1. Here, the architecture and parameters of the pretrained DnCNN are totally copied as initialization for fine-tuning. Fine-tuning takes a model that has already been trained for one given task, and then tunes or tweaks the model to make it perform a second similar task, thus allowing us to take advantage of what the model has already learned without having to develop it from scratch. There are a number of fine-tuning methods according to different scenarios. For example, network pruning is always used to reduce the size of the pretrained model for fine-tuning,

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and thus, we have to choose how many layers we are using, what types of layers we are using, how many nodes we are using in each layer, and add a few new layers and nodes to adapt the new task [42]. However, in many applications, fine-tuning has the same network structure as pretraining, especially when these two learning tasks are very similar or the same. As both pretraining and fine-tuning of our model are to recover HSIs from their noisy observations, we only use the fine-tuning solutions without modifying the DnCNN structure.

B. Discussion

Self-supervised learning in image restoration remains a substantially less developed area, especially for HSI restoration. A few currently existing methods are developed from deep prior, zero-shot learning, and Steins unbiased risk estimate (SURE). Here, we discuss their differences from our method.

Imamura *et al.* proposed a zero-shot HSI denoising method using a 3-D separable image prior (SIP) [43], which automatically creates a training dataset from a single-corrupted HSI and trains a denoising network without any clear images. It is based on the assumption that a network that can recover a clear image x from a corrupted y and also has the ability to recover y from $y' = y + n$, where n represents the extra noise imposed on y. Consequently, the training pair $\{y', y\}$ can be used to replace $\{y, x\}$. Specifically, they added Gaussian noise into an original HSI $\mathcal{Y}$ to create a new noisy HSI $\mathcal{Y}'$, then built a noise-noise pair $\{\mathcal{Y}, \mathcal{Y}'\}$. Finally, a number of HSI patch pairs used for network training can be extracted from $\{\mathcal{Y}, \mathcal{Y}'\}$. The authors further enhanced the performance of the method by replacing the input image with a restored image every a few hundred epochs.

Nguyen *et al.* proposed the SURE-CNN [44] which is an unsupervised CNN as its loss function can be computed only from noisy data. It is because SURE is an unbiased estimation of the mean squared error (MSE) of an estimator, training a CNN using the SURE loss can yield similar results as using the MSE with ground truth in supervised learning. This method is somewhat similar to noise-to-noise learning, but unfortunately, the excessive computation of SURE is infeasible in most cases, so Monte Carlo SURE is used. The SURE loss function for HSI is extended from the idea of [47] and [48] that focuses on RGB color images.

Extended from deep image prior (DIP), Sidorov *et al.* presented the deep hyperspectral prior (DIP-HSI) for HSI denoising, inpainting, and super-resolution [45]. It is based on the conclusion that the structure of the CNN itself encloses the statistical prior of the images. As a result, DIP-HSI trains a 3-D CNN to map a random noise 3-D tensor to a corrupted HSI, and the denoised HSI can be reached before the noise will be recovered during the optimization process. The article [46] is also based on DIP, in which two types of unsupervised DIPs, i.e., 2-D U-Net-like network and 1-D fully connected network, are employed to model the abundance maps and endmembers contained in the HSIs, respectively. In essence, it combines a 2-D network and a 1-D network to accomplish 3-D HSI learning. However, these two methods incline to over-smoothing images since the optimal stopping point of learning is hard to determine.

Similar to our method, all aforementioned three self-supervised learning approaches do not need the clean HSIs used for generating training samples, and instead, they use the pairs between random noise and a corrupted HSI, or between an original corrupted HSI and a new corrupted HSI imposed with further noise, or between two same corrupted HSIs. Nevertheless, our approach uses the neighboring bands of a corrupted HSI to create training pairs. This training data generation paradigm is much simpler and more flexible than the other methods. More importantly, we directly use DNNs originally used for 2-D images as a model. In contrast, the other methods must specially design network architecture for 3-D HSI. Therefore, our method not only simplifies model establishment but also lowers the requirement on the number of training samples.

In summary, the proposed two-stage learning scheme has four distinct properties. First, the fine-tuning samples are selected from the corrupted HSI itself, enabling the learned network to fit the corrupted HSI. Fine-tuning and self-supervised learning are integrated, and thus, the gap in data distribution between training and testing can be reduced. Second, benefitting from the continuous and narrow spectral intervals of hyperspectral imaging, we can consider the underlying clean images of adjacent bands very similar so that the noise-to-noise learning method is used. Third, the spectral correlation between neighboring bands is reflected in noise–noise training pairs, avoiding the design of a specific network architecture for HSIs, making our method adaptable. Finally, as the ordinary DNNs for single-channel images can be used as a backbone network, we have a large range of options on DNN models and a corresponding dataset for pretraining.

IV. Experiments

In this section, we perform experiments on several widely used HSI datasets to validate the proposed denoising method. Moreover, experiments with regard to several important issues, such as the determination of parameters, ablation and experiments, comparison with state-of-the-art approaches, are also conducted.

Experimental Environment: PyTorch 1.6 based on CUDA library is selected as the implemental tools. All experiments are running on a workstation equipped with an NVIDIA Tesla V100 graphics card.

Datasets: Two datasets CAVE and ICVL were imaged by ground-based cameras. Other datasets, i.e., Houston, Kennedy Space Center (KSC), Indian Pines, were captured by airborne-based remote sensing devices so that they cover common scenarios as benchmark datasets.

Compared Methods: Traditional and deep learning methods are selected for comparison, including sparse representation-based method BM3D [10], low-rank representation-based methods LRMR [20], and total-variation-regularized low-rank matrix factorization (LRTV) [47], supervised deep learning-based methods SP3DCNN [25], HSID-CNN [26], HSI-SDeCNN [27], and a self-supervised learning-based

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Fig. 2. Samples of CAVE dataset in pseudocolor.

method SIP [43]. They are commonly used for comparison or newly proposed and are very relevant to our method.

Training Parameters: Our network training is divided into a pretraining stage and a fine-tuning stage. In the pretraining stage, the learning rate is initialized to 10^{-3} in the first 30 epochs then changed to 10^{-4} in the next 70 epochs. In the fine-tuning stage, we set 120 epochs and the learning rate is set to 10^{-5}. Both stages use the Adam optimizer as the gradient descent optimization method and batch size is set to 8. We keep these settings in the experiments on all datasets, which means that the parameters will not be adjusted to overfit any specific dataset.

Noise Setting: We set two types of noises in the experiments, i.e., Gaussian noise and mixed noise. Before adding noise, the magnitudes of pixels in each band are normalized to [0, 255]. In order to better simulate actual situations, we add different levels of noise to clean images. Specifically, in the experiment of Gaussian noise, we add additive white Gaussian noise, and the standard deviation σ of Gaussian noise is taken randomly in a certain range band by band, for example, $\sigma \in [0, 75]$. In the experiment of mixed noise, we add Gaussian noise first, then add impulse noise with 10% intensity to generate simulated noise data.

Evaluation Metrics: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [40], feature similarity index (FSIM) [48], and relative dimensionless global error in synthesis (ERGAS) [49] are used for quantitative assessment. PSNR, SSIM, and FSIM are three conventional picture quality indices (PQIs). ERGAS records the fidelity by computing the weighted sum of MSE and describes the global statistical quality of the restored image. Larger values of PSNR, SSIM, and FSIM imply better performance; meanwhile, a smaller value of ERGAS suggests better performance.

A. Experiments on CAVE

The CAVE dataset [50][1] was gathered by Apogee Alta U260 CCD camera, covering a wide variety of real-world materials and objects. The dataset consists of 32 scenes. Each scene has 31 spectral bands in the range of 400–700 nm and the size of each band image is 512 × 512 pixels. Its pseudocolor image samples are shown in Fig. 2.

A series of experiments on CAVE are conducted to verify the effectiveness of the proposed method. First, we analyze the impacts of three important parameters in our model, the size of neighbor k, the iteration number of fine-tuning, and the weight of loss function α, and determine their suitable values for the subsequent experiments. Second, the ablation experiment is done to show the role of two-stage training. Finally, comparisons with other methods are given to demonstrate the superiority of the proposed method.

[1]https://www.cs.columbia.edu/CAVE/databases/multispectral/

TABLE I

IMPACT OF THE WEIGHT OF LOSS ON PSNR UNDER TWO TYPES OF NOISES ON CAVE DATASET

Noise	α				
	0	0.2	0.5	0.8	1
[0-75]	33.24	**33.27**	33.26	33.22	33.20
mixed	29.76	30.03	30.54	**30.88**	30.87

The original HSIs in CAVE are assumed to be clean or nearly clean so we add noise to simulate the corrupted HSIs. We set up four cases of noises, in which three are Gaussian noise but with different ranges of noise levels, i.e., the σ of a band image is randomly chosen from [0-25], [0-50], and [0-75], respectively, The other one is mixed noise with Gaussian and impulse noise, i.e., adding Gaussian noise with $\sigma \in [0, 75]$ first, then adding impulse noise with 10% intensity to generate noisy HSIs.

The scenes in CAVE are divided into a training set and a testing set. We randomly choose 29 scenes for pretraining and the rest three scenes as the testing samples. In the pretraining stage, we train the DnCNN model in a band-by-band manner using the training set to obtain the pretrained model. In the fine-tuning stage, we perform noise-to-noise learning on each test HSI and predict its denoised HSI at the same time. The quantitative results are the average on all test scenes with five times experiments.

1) Impact of the Weight of Loss: This experiment analyzes the impact of the weights of L_1 and L_2 losses for Gaussian and mixed noises, respectively. Our loss function is based on the combination of L_1 and L_2 norms, defined in (8). We assign different values to α to evaluate the denoising performance, and the results are shown in Table I. Here, the size of neighbor k is set to the default value 3. The best PSNR is obtained when α is set to 0.2 for Gaussian noise and 0.8 for mixed noise. For Gaussian noise, different α has less effect on PSNR. However, for mixed noise, a larger α obviously obtains better results. We attribute this to the fact that L_1 and L_2 losses have their respective properties, especially L_2 is suitable for Gaussian noise; meanwhile, L_1 is more suitable for multiple types of noises, such as impulse noise. If not specified in subsequent experiments, the α is set to 0 for Gaussian noise for the sake of fairness when compared with other approaches and 0.8 for mixed noise only in order to show that noise-to-noise learning can be extended to impulse noise.

2) Iteration Number of Fine-Tuning: The fine-tuning stage is also a denoising procedure, so we try to decrease the training time under the premise of the denoising performance. In the experiment on CAVE with $\sigma \in [0, 25]$, Fig. 3 shows the PSNR and loss change along with the number of fine-tuning epochs, from which it is found that after about 120 epochs both of PSNR and loss tend to oscillatory stationary state.

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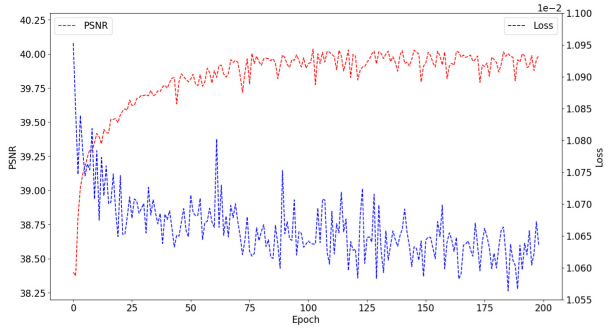


Fig. 3. PSNR and loss change along with the number of fine-tuning epochs.

TABLE II
IMPACT OF THE SIZE OF NEIGHBOR k ON PSNR UNDER TWO TYPES OF NOISES IN CAVE DATASET

Noise	k				
	1	2	3	4	5
[0-75]	33.27	33.5	**33.62**	33.6	33.55
mixed	30.88	30.86	**30.98**	30.91	29.82

We also find this phenomenon in other experiments in our article; therefore, we stop fine-tuning after 120 epochs in all experiments. In practice, this parameter is not difficult to be determined, as an approximate optimal denoising performance can be achieved as long as a bit large number of epochs is set.

3) Impact of the Size of Neighborhood: In the fine-tuning stage, we use the test HSI itself as the source of training samples by a self-supervised learning scheme. This parameter determines the number of training pairs, i.e., the larger size the more training samples, but the assumption that a training pair has the same underlying clean image might be violated more or less, which is a tradeoff problem. Two neighboring bands are selected as the input and target of a training pair, i.e., if the ith band is chosen as input, the jth band is its target if $1 \leq \|i - j\| \leq k$. Table II shows the denoising results with different sizes k under Gaussian noise and mixed noise. It can be found that as k increases, PSNR first rises and then falls, and the best result is achieved when $k = 3$ under both Gaussian noise and mixed noise. If not specified, we set $k = 3$ in all subsequent experiments.

4) Different Fine-Tuning Strategies: We only use the fine-tuning strategies without modifying network structure. Here, four popular fine-tuning strategies are considered: S1—using the pretrained network as initialization, and then fine-tuning the whole network with new training samples; S2—using the pretrained network as initialization, then only fine-tuning the last layer, and freezing all other layers; S3—using the pretrained network as initialization but the last layer is randomly initialized, then only fine-tuning the last layer, and freezing all other layers; S4—using the pretrained network as initialization, then first fine-tuning the last layer and freezing all other layers, and second, fine-tuning the parameters in all layers. Table III shows the denoising results with different fine-tuning strategies under Gaussian noise and mixed noise on the CAVE dataset. It can be found that the results of fine-tuning all parameters (S1 and S4) are better than those of only fine-tuning the parameters of the last layer (S2 and S3). The S4 is very slightly better than S1, but its number of epochs in fine-tuning is about times of S1, due to its two-step training. Therefore, in all other experiments, we use S1 as a fine-tuning solution for computation and performance purposes.

TABLE III
DENOISING RESULTS ON CAVE WITH DIFFERENT FINE-TUNING SOLUTIONS

Noise	S1	S2	S3	S4
[0-25]	39.95	38.64	38.54	**39.96**
[0-50]	**34.98**	33.74	33.65	34.97
[0-75]	33.55	32.28	32.26	**33.75**
mixed	30.46	29.75	29.59	**30.52**

TABLE IV
ABLATION STUDY OF TWO STAGES (I AND II) WITH PSNR ON CAVE DATASET UNDER FOUR TYPES OF NOISES

Methods	I	II	[0-25]	[0-50]	[0-75]	mixed
Noise	-	-	31.24	24.13	23.72	12.41
Pre-trained	✓	-	38.4	32.92	31.94	28.74
Fine-tuned w.o.	-	✓	38.02	33.24	31.97	26.07
Fine-tuned w.t.	✓	✓	40.04	35.07	33.62	30.39

5) Ablation Study of Two-Stages Training: This ablation study is to show the contributions of pretraining and fine-tuning in our method, respectively. Table IV displays PSNR of the test HSIs under four noise cases. The *pretrained* DnCNN is learned using the CAVE training set and is considered as the baseline in this experiment. If we only conduct the fine-tuning stage without pretraining (*fine-tuned w.o.*), the PSNR will be slightly lower than *pretrained*; that is because the number of samples used for fine-tuning is much smaller than that for pretraining. The results are significantly improved when both stages are used (*fine-tuned w.t.*). Fig. 4 displays more detailed results. We choose one of the test scenes "chart and stuffed toy" to show the PSNRs of *pretrained*, *fine-tuned w.o.*, and *fine-tuned w.t.*, respectively, for each band, from which we can find that the improvement of the two-stage scheme is reflected on all bands.

6) Comparative Experiment: In this experiment, we compared the proposed two-stage denoising method with some state-of-the-art approaches. Table V shows the quantitative results. In most cases, our method can achieve the best results. Figs. 5 and 6 are visual results in CAVE with Gaussian noise and mixed noise. BM3D and LRMR cannot remove all the noises. LRTV successfully removes the noise, but images become a little bit over smoothed and some edge information is lost. SP3DCNN, HSID-CNN, DnCNN, and SIP lose some local structures. HSI-SDeCNN provides appealing visual results in the case of Gaussian noise, but it performs poorly under mixed noise. Our results successfully remove two types of noises and preserve the fine structures of HSIs.

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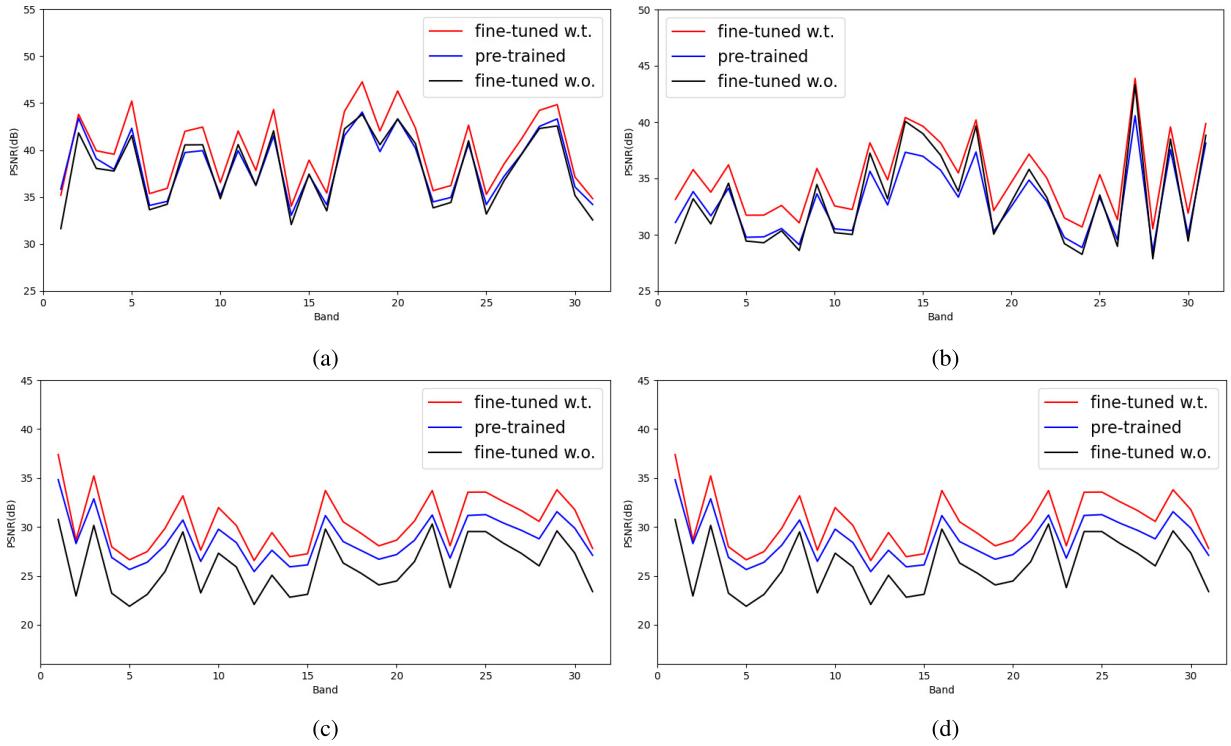


Fig. 4. Results (PSNR) of each band in the *chart and stuffed toy* HSI in CAVE under four noise cases. (a) Gaussian noise with $\sigma \in [0, 25]$. (b) Gaussian noise with $\sigma \in [0, 50]$. (c) Gaussian noise $\sigma \in [0, 75]$. (d) Mixed noise.

TABLE V
COMPARISON OF DIFFERENT METHODS ON CAVE DATASET

Noise	Index	Methods									
		Noisy	BM3D [10]	LRMR [20]	LRTV [47]	SP3D CNN[25]	HSI-SDe CNN[27]	HSID-CNN[26]	DnCNN [20]	SIP[43]	Ours
[0-25]	PSNR	31.24	35.98	34.78	34.71	34.02	39.62	34.33	38.4	34.78	**39.95**
	SSIM	0.5958	0.8503	0.7427	0.962	0.9353	0.9603	0.9504	0.9501	0.9428	**0.9624**
	FSIM	0.9113	0.9745	0.9504	0.9816	0.9724	**0.986**	0.9781	0.9837	0.9737	0.9855
	ERGAS	143.12	72.09	92.13	73.17	78.78	43.56	77.36	50.78	72.66	**43.53**
[0-50]	PSNR	24.13	29.83	28.76	32.42	31.26	**36.18**	32.24	32.92	34.28	34.98
	SSIM	0.3329	0.6831	0.5268	0.9311	0.892	0.9082	0.903	0.823	0.8958	**0.9175**
	FSIM	0.7762	0.9381	0.8827	0.9616	0.9463	0.9658	0.9531	0.9498	0.9613	**0.9661**
	ERGAS	320.42	148.14	177.59	95.64	108.02	**73.71**	99.03	94.58	77.1	74.53
[0-75]	PSNR	23.72	28.6	27.58	31.86	30.93	32.81	31.65	31.94	30.83	**33.55**
	SSIM	0.3192	0.6541	0.4995	**0.9179**	0.7749	0.7532	0.8774	0.838	0.7718	0.8805
	FSIM	0.748	0.9283	0.8519	0.9555	0.896	**0.9567**	0.9434	0.9425	0.9342	0.9502
	ERGAS	380.22	175.08	245.33	102.24	115.76	91.93	107.3	104.72	114.02	**88.25**
mixed	PSNR	12.41	20.51	23.57	30.21	28.73	27.79	29.03	28.74	26.69	**30.46**
	SSIM	0.0786	0.2863	0.4168	0.7655	0.7145	0.7273	0.7948	0.7189	0.6594	**0.8076**
	FSIM	0.4871	0.7454	0.7807	0.8779	0.8596	0.9086	0.905	0.9055	0.876	**0.9217**
	ERGAS	948.28	404.84	484.97	169.66	146.03	160.99	143.15	148.6	182.31	**123.89**

B. Experiments on ICVL

To further validate our method, we use a larger dataset ICVL [51],[2] which was acquired on the ground using a Specim PS Kappa DX4 hyperspectral camera. ICVL consists 201 scenes collected at 1392 × 1300 spatial resolution with 31 spectral bands. Some pseudo color images in this dataset are shown in Fig. 7.

We choose 150 HSIs for training and the rest HSIs are treated as the testing set. We crop multiple overlapped patches with the spatial size of 512 × 512 from each training HSI for

[2]http://icvl.cs.bgu.ac.il/hyperspectral/

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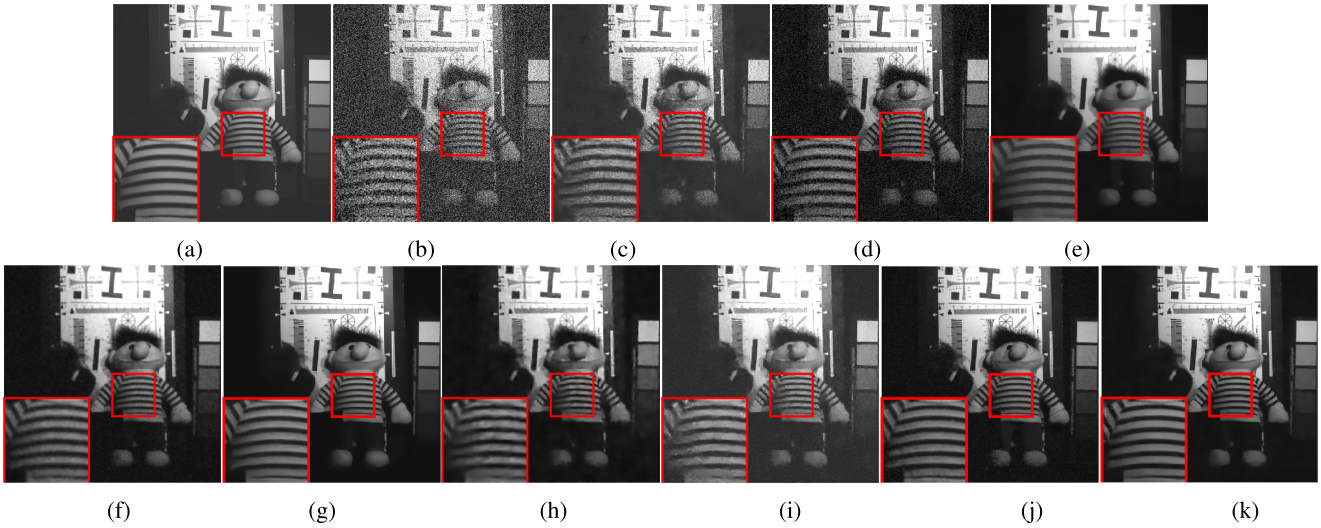


Fig. 5. Results at the 26th band of the *chart and stuffed toy* HSI in CAVE with Gaussian noise, $\sigma \in [0, 75]$. (a) Clean. (b) Noisy. (c) BM3D. (d) LRMR. (e) LRTV. (f) SP3DCNN. (g) HSI-SDeCNN. (h) HSID-CNN. (i) DnCNN. (j) SIP. (k) Ours.

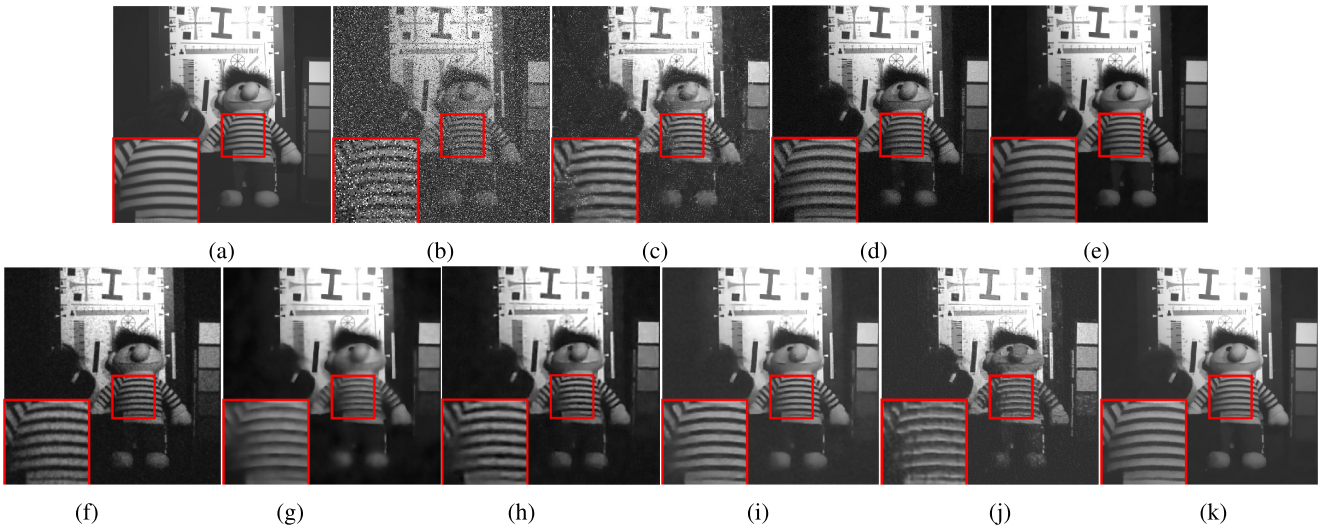


Fig. 6. Results at the 27th band of the *chart and stuffed toy* HSI in CAVE with mixed noise. (a) Clean. (b) Noisy. (c) BM3D. (d) LRMR. (e) LRTV. (f) SP3DCNN. (g) HSI-SDeCNN. (h) HSID-CNN. (i) DnCNN. (j) SIP. (k) Ours.

the pretraining stage. As for the testing set, we cropped out the central region of each image with the same spatial size of 512×512 for evaluation. Similar to the experiments in CAVE, we assume the original HSIs in ICVL are clean and set up four cases of noises, including three cases of adding different levels of Gaussian noises with $\sigma \in [0, 15], [0, 55], [0, 95]$, respectively, and one case of adding mixed noise of Gaussian noise with $\sigma \in [0, 95]$ and impulse noise with 10% intensity.

Table VI shows the comparative results of all methods on four cases. Overall, our method performs much better than other compared methods. Figs. 8 and 9 display the visual results in Gaussian noise and mixed noise cases.

In order to evaluate the computational load of our proposed approach, we compare its running time with other compared methods. The result is shown in Table VII. The running time for all learning-based methods does not include training time, and for our method, the pretraining time is not included.



Fig. 7. Samples of ICVL dataset in pseudocolor.

It can be found that our method spends more time than traditional methods and much more than HSID-CNN and DnCNN, due to that the fine-tuning stage in our method contains the training and test simultaneously; thus, the time

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TABLE VI
COMPARISON OF DIFFERENT METHODS ON ICVL DATASET

Noise	Index	Methods									
		Noisy	BM3D [10]	LRMR [20]	LRTV [47]	SP3D CNN[25]	HSI-SDe CNN[27]	HSID-CNN[26]	DnCNN [20]	SIP[43]	Ours
[0-15]	PSNR	33.09	40.32	38.67	42.96	40.25	43.09	43.31	44.08	41.28	**44.29**
	SSIM	0.6205	0.8737	0.8659	0.9584	0.9597	0.971	0.9637	0.9687	0.97	**0.9706**
	FSIM	0.9268	0.9698	0.9833	0.9886	0.9805	**0.9868**	0.9798	0.9817	0.985	0.9828
	ERGAS	139.47	56.89	39.19	22.98	46.65	30.62	36.88	30.45	37.71	**29.29**
[0-55]	PSNR	23.1	32.4822	30.31	36.08	36.15	37.02	36.85	36.9	35.55	**38.08**
	SSIM	0.2624	0.6721	0.6196	0.862	0.8976	0.917	0.9121	0.88	0.9088	**0.9181**
	FSIM	0.7343	0.8782	0.9298	**0.9449**	0.9456	0.9433	0.9346	0.9385	0.941	0.9428
	ERGAS	422.85	179.28	103.35	**58.09**	68.14	63.81	68.21	70.64	74.21	60.54
[0-95]	PSNR	17.43	29.72	27.58	33.21	32.99	35.32	34.64	33.91	32.94	**35.34**
	SSIM	0.1547	0.5948	0.5367	0.7999	0.7963	0.82	**0.8856**	0.8068	0.8676	0.8676
	FSIM	0.5649	0.8233	0.8736	0.8895	0.8947	0.9164	0.9081	0.9077	0.9007	**0.9171**
	ERGAS	880.59	248.46	184.75	109.27	97.98	84.86	87.08	98.47	100.68	**82.62**
mixed	PSNR	9.17	26.45	23.3	29.77	30.26	30.87	33.26	32.33	29.64	**33.75**
	SSIM	0.0293	0.5625	0.4063	0.758	0.7207	0.7861	0.855	0.7652	0.7753	**0.8573**
	FSIM	0.3573	0.7783	0.8018	0.8578	0.8758	0.894	0.8961	0.8941	0.8697	**0.9045**
	ERGAS	1768.65	279.1	352.33	149.98	133.29	125.3	100.81	113.46	150.12	**96.17**

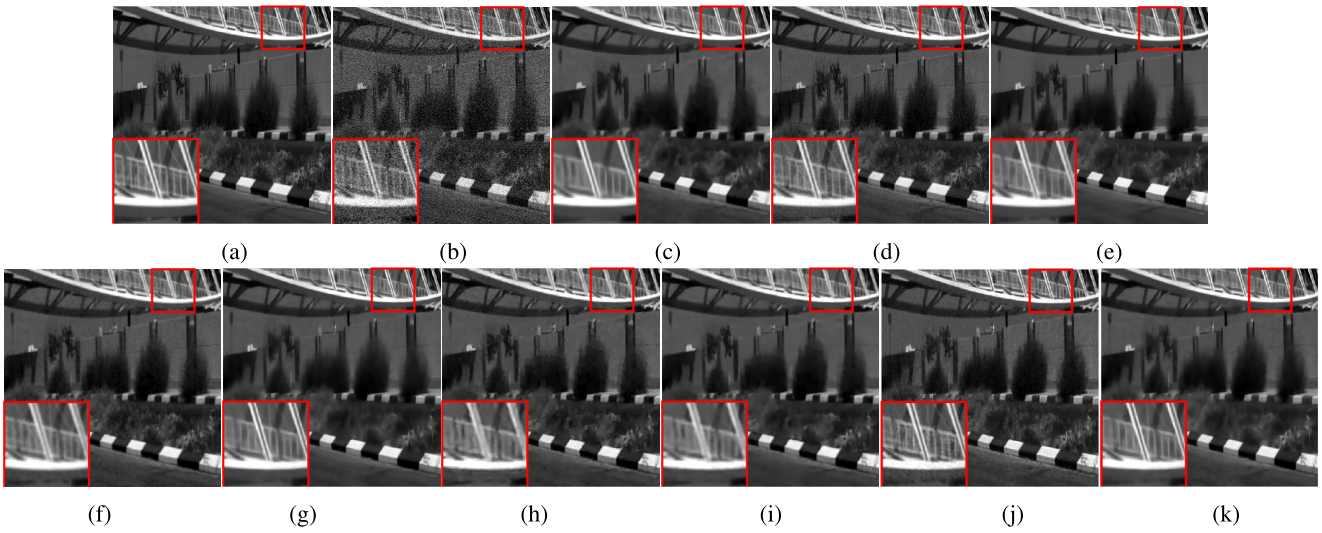


Fig. 8. Results at the 19th band of the *gavyam 0823-0933* HSI in ICVL with Gaussian noise, $\sigma \in [0, 95]$. (a) Clean. (b) Noisy. (c) BM3D. (d) LRMR. (e) LRTV. (f) SP3DCNN. (g) HSI-SDeCNN. (h) HSID-CNN. (i) DnCNN. (j) SIP. (k) Ours.

load of training is relatively expensive. However, our method is very faster than SIP that is also a self-supervised method, and needs training when denoising an HSI. As for memory complexity, our method is fully dependent on the 2-D DNN model. In this article, we use DnCNN as a basic model; therefore, its memory complexity equals DnCNN, whose the number of parameters is about 39k.

C. Experiments on Houston and KSC Datasets

The Houston dataset[3] was released by the Hyperspectral Image Analysis Laboratory and the National Center for Airborne Laser Mapping (NCALM) at the University of Houston, Houston, TX, USA. The dataset includes multispectral-LiDAR, hyperspectral, and very high-resolution RGB images, acquired over the University of Houston campus and its neighborhood on February 16, 2017. We only use its HSI data, which contains 50 bands ranging from 380 to 1050 nm. The spatial size of each band is 2384 × 601. Its pseudocolor image is shown in Fig. 10. Another dataset is KSC,[4] which was acquired by the NASA AVIRIS sensor over the KSC, Merritt Island, FL, USA, in 1996. The original data contain 224 bands of 10-nm width from 400 to 2500 nm and has 512 × 614 pixels

[3]https://hyperspectral.ee.uh.edu/?page_id=1075

[4]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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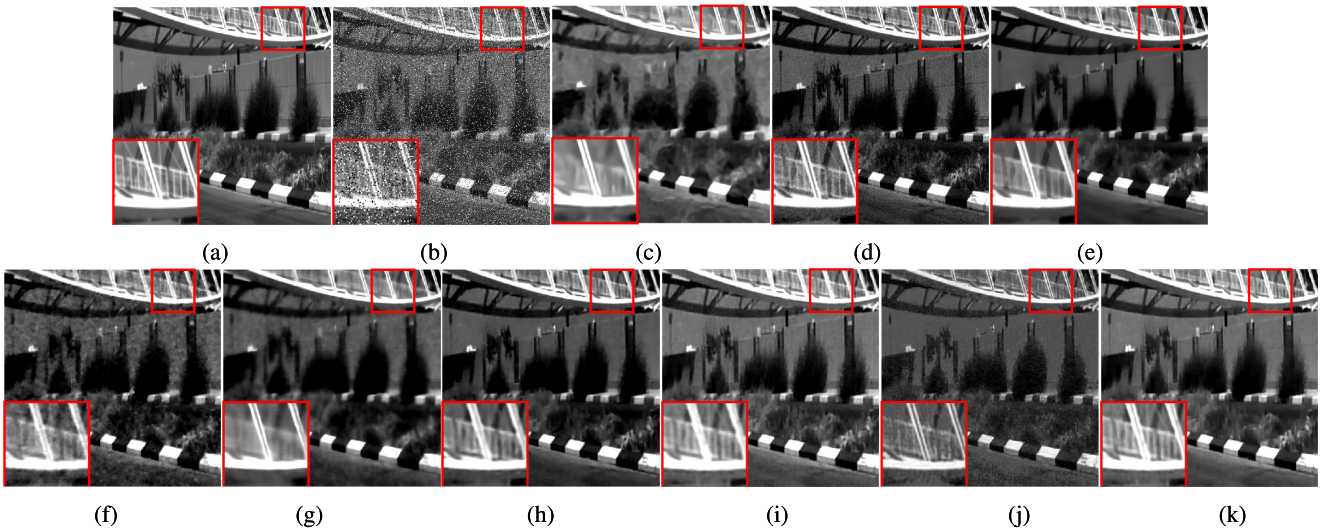


Fig. 9. Results at 22th band of the *gavyam 0823-0933* HSI in ICVL with mixed noise. (a) Clean. (b) Noisy. (c) BM3D. (d) LRMR. (e) LRTV. (f) SP3DCNN. (g) HSI-SDeCNN. (h) HSID-CNN. (i) DnCNN. (j) SIP. (k) Ours.

TABLE VII
RUNTIME OF DIFFERENT METHODS ON ICVL DATASET

Methods	BM3D	LRMR	LRTV	SP3DCNN	HSI-SDeCNN	HISD-CNN	DnCNN	SIP	Our
Times(s)	212.22	181.38	271.17	1.99	2.4	1.06	2.88	2259.62	484.33

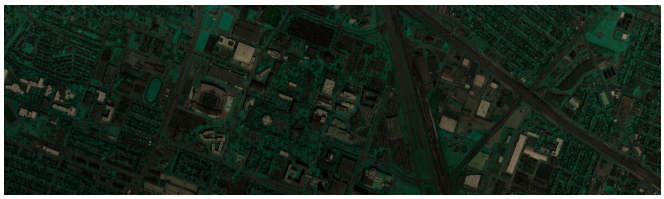


Fig. 10. Pseudocolor image generated by bands 20, 30, and 40 in Houston dataset.

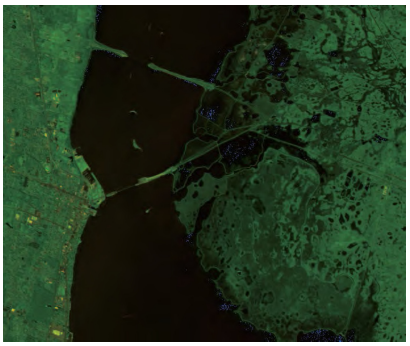


Fig. 11. Pseudocolor image generated by bands 5, 50, and 150 in KSC dataset.

in each band. After removing water absorption and low SNR bands, 176 bands are used for our experiment. Fig. 11 shows its pseudocolor image.

In the experiments, these two original HSIs are also assumed to be clean. The same as experiments on the CAVE and ICVL datasets, two cases of added noise are used for generating corrupted HSIs, i.e., Gaussian noise with $\sigma \in [0, 75]$ and

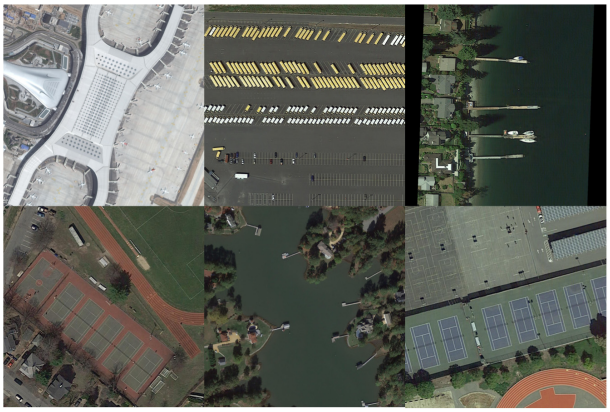


Fig. 12. Some samples in DOTA dataset.

mixed noise of Gaussian noise with $\sigma \in [0, 75]$ and impulse noise with 10% intensity. In order to simplify the experiments, we crop sub-HSIs with the spatial size of 512×512 from two original HSIs as testing samples.

Since the Houston and KSC datasets are independent and each has just one HSI, we only use them as testing HSIs. Through the experiments in this section, we demonstrate the distinct advantage of the proposed two-stage learning method, i.e., our backbone network DnCNN is originally designed for 2-D images; therefore, all grayscale, color, multi/HSI images can be used as training samples for the pretraining stage, which is different from most of DNN-based HSI denoising methods whose training samples are required to be HSIs or HSI patches. This feature enables our method to be more flexible and usable in practice.

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TABLE VIII
RESULTS ON HOUSTON AND KSC DATASETS

Data	Noise	Index	Methods									
			Noisy	BM3D [10]	LRMR [20]	LRTV [47]	SP3DC NN[25]	HSI-SDe CNN[27]	HSID-CNN[26]	DnCNN [20]	SIP[43]	Ours
Houston	[0-75]	PSNR	20.46	27.98	28.4	34.24	33.26	33.49	33.32	33.18	31.68	**34.73**
		SSIM	0.1899	0.5595	0.5343	**0.875**	0.8399	0.8194	0.7548	0.7129	0.8491	0.8495
		FSIM	0.6431	0.8008	0.8784	0.8759	0.8984	**0.9435**	0.9036	0.9059	0.9217	0.9014
		ERGAS	729.73	328.88	201.46	94.98	102.7	100.96	98.88	121.42	119.89	**96.33**
	mixed	PSNR	11.01	26.73	28.11	33.47	30.09	30.32	32.33	31.55	27.86	**33.76**
		SSIM	0.0335	0.611	0.5759	0.7926	0.7815	0.8241	0.7496	0.6887	0.7433	**0.8358**
		FSIM	0.4461	0.786	0.854	0.8843	0.8992	0.9013	0.8931	0.9036	0.8698	**0.9062**
		ERGAS	1373.01	328.39	302.09	214.52	148.09	146.96	123.44	130.28	189.77	**99.74**
KSC	[0-75]	PSNR	19.83	27.1	29.79	30.89	31.3	31.55	30.71	31.05	31.29	**31.74**
		SSIM	0.2444	0.5904	0.6372	**0.7999**	0.7922	0.7944	0.7759	0.715	0.7737	0.7868
		FSIM	0.7016	0.8089	0.9004	0.8378	0.8971	0.8868	0.8901	0.9024	**0.9333**	0.9047
		ERGAS	952.29	370.43	250.79	255.09	237.02	202.24	240.15	250.15	**161.84**	233.08
	mixed	PSNR	9.55	23.26	27.27	30.17	27.99	29.5	29.14	28.74	26.79	**30.33**
		SSIM	0.0381	0.4256	0.5585	**0.7774**	0.6565	0.7545	0.7192	0.587	0.6514	0.7511
		FSIM	0.4374	0.7372	0.8821	0.8102	0.8651	0.9034	0.8373	0.8641	0.8859	**0.9088**
		ERGAS	2017.51	467.19	295.23	266.77	330.74	278.1	263.95	287.23	283.78	**251.11**

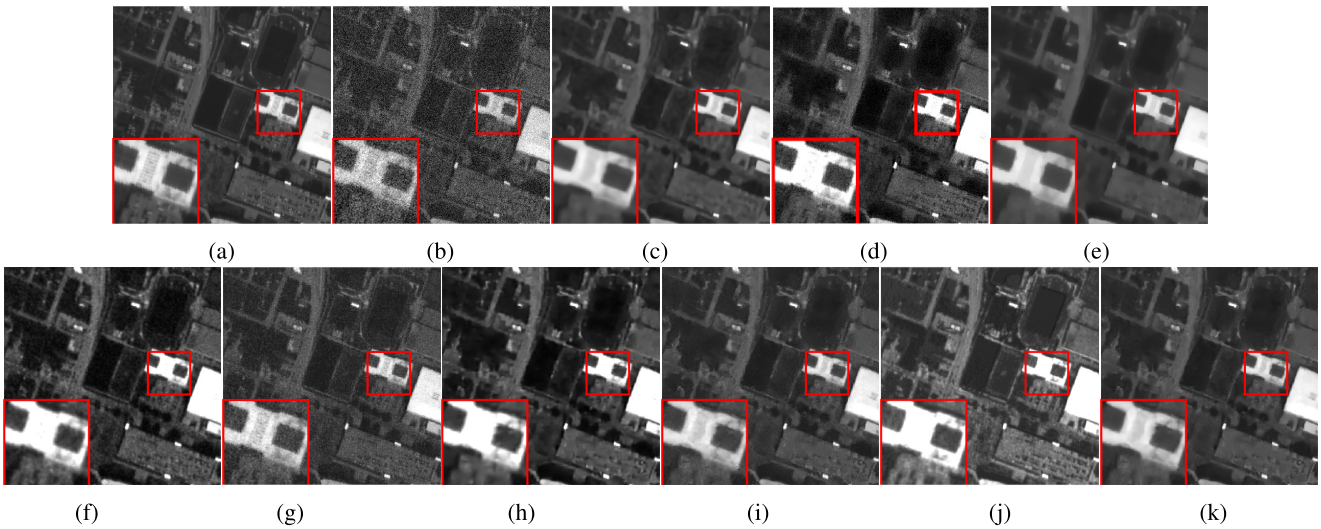


Fig. 13. Results at the eighth band of Houston dataset under Gaussian noise, $\sigma \in [0, 75]$. (a) Clean. (b) Noisy. (c) BM3D. (d) LRMR. (e) LRTV. (f) SP3DCNN. (g) HSI-SDeCNN. (h) HSID-CNN. (i) DnCNN. (j) SIP. (k) Ours.

Here, we choose the DOTA dataset [52][5] as the external dataset for pretraining. DOTA dataset contains 2806 aerial RGB or grayscale images from different sensors and platforms. The size of each image is 4000 × 4000 pixels, and a wide variety of objects are included in these images. Fig. 12 shows some image samples from this dataset. We crop the central region of each image in DOTA with the size of 512 × 512 as the training data. We train a single-channel DnCNN using the DOTA dataset to obtain the pretrained model, then fine-tune the pretrained model using the Houston and KSC dataset, respectively, and finally get their denoised HSIs.

Similar to the Houston and KSC datasets, DOTA was also captured from optical imaging devices of remote sensing, and it has some similar scenes to Houston and KSC; meanwhile, DOTA has not any HSI, and therefore, it is very suitable for evaluating the property of easy-to-use of our method in the collection of training samples. Since three compared DNN-based methods, SP3DCNN, HSID-CNN, and HSI-SDeCNN, must be trained with HSI samples, they are learned with all scenes in the CAVE and ICVL datasets. On the contrary, the traditional model-based methods BM3D, LRMR, LRTV, and self-supervised learning-based method SIP, do not need an external training set.

Table VIII shows the experimental results. It can be found that our method has superior overall performance compared with other approaches. Compared with SP3DCNN, HSID-CNN, and HSI-SDeCNN that need a collection of HSIs as training samples, our method not only has a better performance but also does not need HSIs as training samples. The visual results in the Gaussian noise case are shown in Figs. 13 and 14, in which our method is able to well remove the real noise while preserving the image details of the local structure.

[5]https://captain-whu.github.io/DOTA/dataset.html

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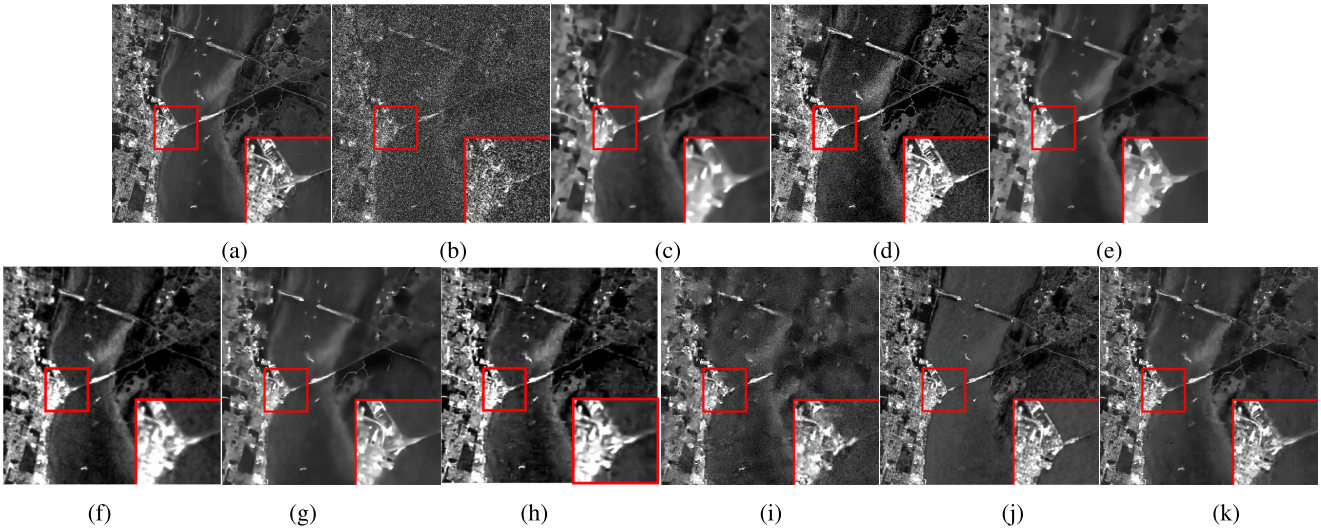


Fig. 14. Results at the 20th band of KSC dataset under Gaussian noise, $\sigma \in [0, 75]$. (a) Clean. (b) Noisy. (c) BM3D. (d) LRMR. (e) LRTV. (f) SP3DCNN. (g) HSI-SDeCNN. (h) HSID-CNN. (i) DnCNN. (j) SIP. (k) Ours.

TABLE IX
CLASSIFICATION RESULTS WITH DENOISED INDIAN PINE DATASET

	Noisy	BM3D	LRMR	LRTV	SP3DCNN	HSI-SDe CNN	HSID-CNN	DnCNN	SIP	Our
OA %	53.58	82.16	74.37	76.28	78.03	81.28	81.73	80.5	73.73	**82.35**
Kappa	0.4675	0.7897	0.7013	0.7207	0.7619	0.7731	0.7871	0.7701	0.7071	**0.7917**

D. Experiments on Indian Pines Dataset

All previous experiments are simulated, as their noise is artificially added. In this experiment, we evaluate our method on the real-world noisy HSI dataset—Indian Pines.[6] This dataset was collected by AVIRIS over Northwest Indiana, USA, in 1992. It has 220 bands, in which different bands have different levels of noise. Slight differences in noise levels can even be observable at different locations in the same band image. A part of this scene with the spatial size of 512×512 is selected. Its 118th and 149th band images are shown in Fig. 15, from which we can find that their noise levels are very different. This is due to the physical mechanism of its imaging sensor and the spectral properties of surface materials. Furthermore, its noise types are not limited to Gaussian noise; therefore, it is a real and complicated case.

The pretraining stage is the same as the experiments in Section IV-C, i.e., we use the DOTA dataset as training samples, and their corrupted images are generated by randomly adding Gaussian noise with $\sigma \in [0, 75]$. And the settings of other methods are also the same as the experiments in Section IV-C. Fig. 16 shows the visual results of denoising, in which our method produces a clearer image than others. This experiment demonstrates that the proposed method can work well in real situations.

As this dataset provides the ground-truth labels of 37 classes for classification, we evaluate the denoising performance through classification tasks. The support vector machine (SVM) with linear kernel function is adopted as a classifier for simplification, and the spectral signature of the pixel is used as a feature vector. There are all 262 144 labeled pixels in this area. We randomly select 150 samples/pixels for every class as a training set and the rest of the labeled samples as a test set. The classification results are shown in Table IX, including the results before and after applying the different denoising methods separately. It also demonstrates that our denoised HSI leads to the best classification performance.

[6]https://purr.purdue.edu/publications/1947/1

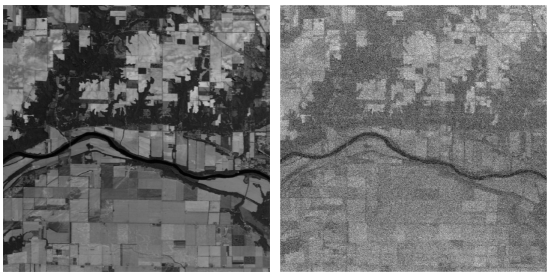


Fig. 15. Two bands of Indian Pines dataset. (a) 118th band and (b) 149th band.

E. Experiments on Urban Dataset

This experiment also runs on a real-world noisy HSI dataset—Urban, which has more complex material distributions than Indian Pines Dataset. The Urban dataset is recorded by the HYDICE sensor in October 1995, which has 210 spectral bands ranging from 400 to 2500 nm and 307×307 pixels

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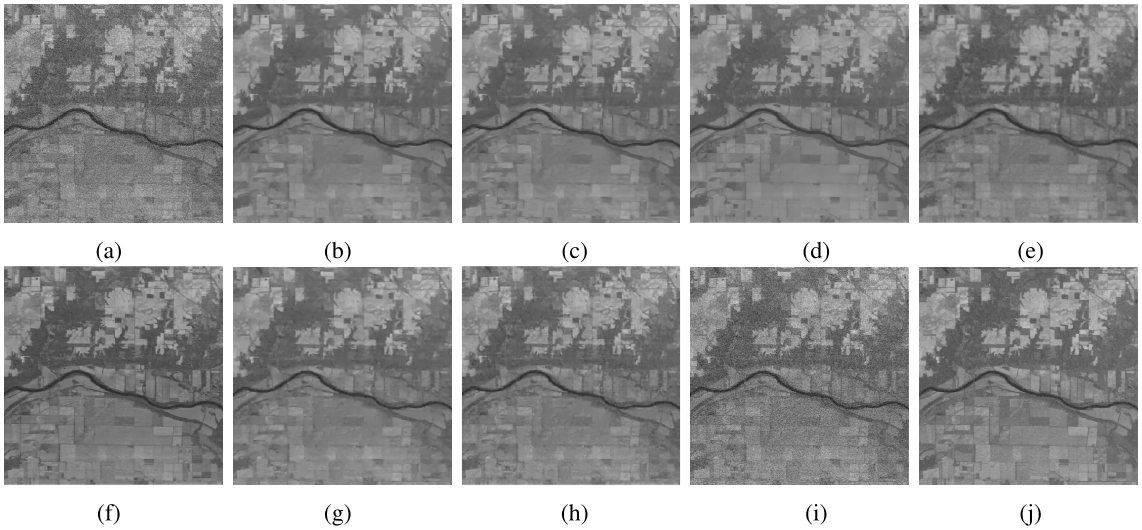


Fig. 16. Results at the 149th band of Indian Pines dataset. (a) Orig. (b) BM3D. (c) LRMR. (d) LRTV. (e) SP3DCNN. (f) HSI-SDeCNN. (g) HSID-CNN. (h) DnCNN. (i) SIP. (j) Ours.

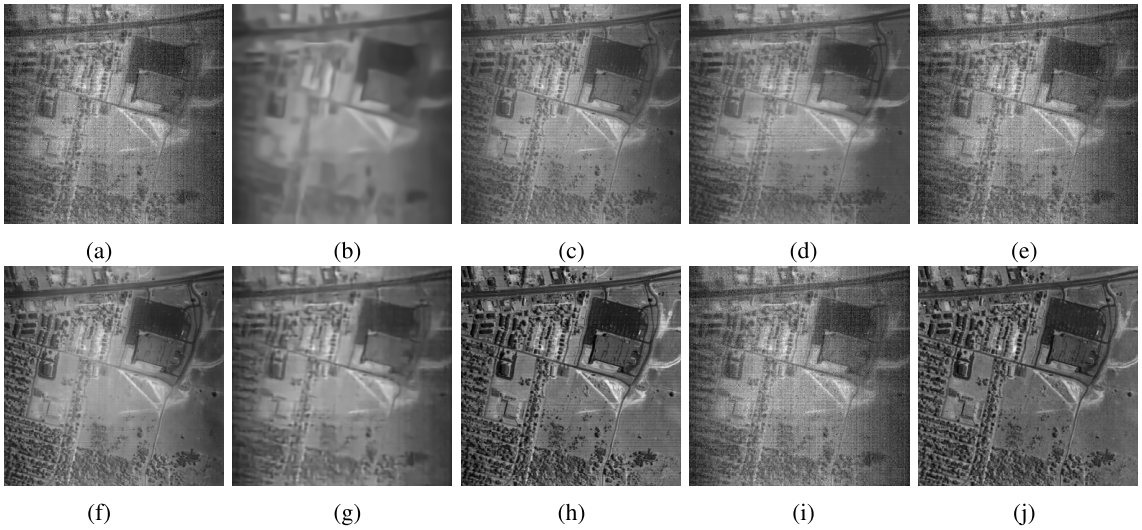


Fig. 17. Results at the 104th band of Urban dataset. (a) Noisy. (b) BM3D. (c) LRMR. (d) LRTV. (e) SP3DCNN. (f) HSI-SDeCNN. (g) HSID-CNN. (h) DnCNN. (i) SIP. (j) Ours.

for each band. We crop part of spatial size 300 $\times$ 300 for experiments. Other experimental settings are the same as for the experiments on the Indian Pine dataset. Fig. 17 shows the visual denoising results on the 104th band with different approaches, in which our method produces a clearer image than others and also retains more details to avoid overly smooth effects.

V. Conclusion

In this article, we propose a two-stage deep learning method for HSI restoration from noisy observations. Through the combination of pretraining and fine-tuning stages, the knowledge of image structure learned from external training samples and the target HSI need to be denoised can be fully exploited so that a competitive performance of HSI restoration is achieved compared with the state-of-the-art approaches. Its outstanding feature is the capability of using denoising DNNs originally designed for ordinary grayscale or RGB color images, such as DnCNN, to solve the 3-D HSI denoising problem. Since we do not need to purposely design network architecture for the spectral–spatial hybrid structure of HSIs, an external training set of HSIs is not mandatory and various types of optical images can be used as training samples, which overcomes the difficulty of acquiring training samples in deep learning. Alternatively, we take the spectral correlation between bands of an HSI into account by noise-to-noise learning and self-supervised learning schemes instead of by network architecture, which not only integrates the learning and denoising tasks into a consistent frame but also only use the corrupted HSI to be denoised to generate training samples. Furthermore, the corrupted HSI to be denoised is involved in the learning procedure; therefore, its noise type and level are also reflected in the learned network. As a result, our method can be applied to more general situations beyond Gaussian noise, e.g., mixed noise and real-world complex noise, although the pretraining is done under Gaussian noise.

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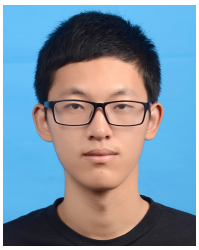
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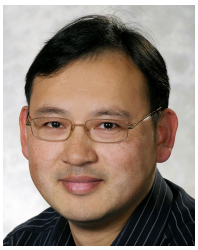
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